

System conditions for offshore electricity storage

A system-level assessment of cost-effectiveness and system impacts in a high offshore
wind Dutch power system

Marijn van Tienhoven

5109361 | mvantienhoven@tudelft.nl

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Supervisors:

Prof. dr. ir. M. A. (Machteld) van den Broek – TU Delft (First Supervisor)

Dr. E. (Enno) Schröder – TU Delft (Second Supervisor)

ir. Bas Janssen – RWE (External Supervisor)

Dr. F. (Francesco) Lombardi, PhD – TU Delft (Advisor)

Abstract

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1 Introduction

1.1 Societal relevance and problem definition

Climate change, triggered by decades of accumulation of greenhouse gas emissions, presents an enormous challenge. The European Commission and the European Council launched the European Green Deal in December 2019 to limit global temperature rise to $+1.5^{\circ}\text{C}$ relative to pre-industrial levels (European Council, 2024). The strategy sets a path for a green transition in the European Union (EU) to achieve climate neutrality by 2050 (European Commission, 2021).

Offshore wind energy is a key pillar of the future low-carbon European power system, due to its high resource potential, relatively stable production profiles, and limited competition for land. The European Commission has set a target of installing 60 GW of offshore wind capacity by 2030 and 300 GW in 2050 (European Commission, 2020). The Netherlands alone plans to have 21 GW of offshore wind capacity in 2030 and 70 GW by 2050, although recent setbacks driven by high construction costs, supply chain bottlenecks, and grid congestion will likely delay short-term goals. Still, to put the ambitions into perspective, the Dutch transmission system operator TenneT estimates that in 2030 offshore wind could generate roughly 75% of Dutch electricity demand, underscoring its scale and strategic importance for achieving climate neutrality (TenneT, 2025b).

However, the large-scale expansion of offshore wind also points to fundamental challenges in transforming energy systems. Electricity generation will increasingly be concentrated offshore, while demand remains onshore (ENTSO-E, 2025b). Offshore wind is connected through offshore substations, export cables and landing points to the onshore grid. Suitable locations for these landing points are limited due to coastal space constraints, environmental restrictions, and the need for nearby high-voltage grid infrastructure that can integrate large volumes of power (Ministerie van Economische Zaken en Klimaat, 2022). Behind the landing points, electricity is transported via the inland transmission network to reach demand. These landing points and the inland grid behind them could become structural bottlenecks as offshore capacity grows, because they have to transport increasing volumes of power (ENTSO-E, 2025b).

Next to this geographical mismatch, the variability of wind generation also leads to a temporal mismatch between supply and demand. Periods of little to no wind generation require flexibility from demand response, dispatchable generation, imports, or storage to maintain reliability (ENTSO-E, 2025a; TNO, 2022). In addition, the variability also affects how infrastructure can be used. In times of high wind output, generation may exceed the capacity of export cables, landing points, and downstream transmission networks, resulting in congestion, curtailment of renewable generation and/or the need for grid reinforcement (Angizeh et al., 2023; ENTSO-E, 2025a). Conversely, during periods of no (or low) production, transmission infrastructure is underutilized, reducing the cost-effectiveness.

Energy storage is widely highlighted as a key solution to address these challenges in systems with a high share of renewable energy (Angizeh et al., 2023; Qazi et al., 2025; Wang et al., 2019; Wiegner, 2025). During periods of high wind power output, storage can absorb excess generation and release energy during periods of shortage. By shifting electricity (peaks) over time, supply and demand can still be matched. In a future system with a much larger share of wind generation, the role of such flexibility becomes increasingly important, raising

the question of how and where storage should be integrated within the power system (Qazi et al., 2025; Rabanal et al., 2024).

Integration of energy storage directly coupled to offshore wind farms is receiving increasing attention in the literature. While offshore storage is still in a developing phase, recent studies highlight its potential to reshape offshore wind integration (Atherton et al., 2023; Juhlin, Eren, et al., 2024; Li & DeCarolus, 2015; Wang et al., 2019; Wiegner et al., 2024). By storing energy offshore, peak loading of export cables and landing points can be reduced. This may reduce the need for grid reinforcements, could lower curtailment of wind energy at the source, and might even reduce pressure on landing point capacity requirements (Atherton et al., 2023; Rabanal et al., 2024). Furthermore, both offshore and onshore storage may relieve pressure on the downstream onshore grid behind the landing points. In some regions, however, spatial constraints may limit large-scale onshore storage deployment near landing points (Angizeh et al., 2023; Ministerie van Economische Zaken en Klimaat, 2022). On the other hand, offshore storage faces higher installation and maintenance costs, marine operational risks, and greater regulatory uncertainty compared to onshore storage (Bijl, 2019; TNO, 2022; Wang et al., 2019).

The characteristics and system impacts of offshore storage can thus be fundamentally different from those of onshore storage. Although offshore storage is unlikely to fully replace onshore storage, offshore storage may provide additional system value under certain conditions where spatial constraints, transmission infrastructure, and variability of generation interact (Rabanal et al., 2024; Wiegner et al., 2024). Understanding these conditions is societally relevant, because decisions on offshore infrastructure, onshore grid development, storage deployment and in general the way offshore wind is integrated, shape the design of the future energy system and partly determine the success of the energy transition (ENTSO-E, 2025b; TenneT, 2025b).

1.2 Literature on offshore storage

Academic literature broadly recognizes the potential role of storage in a future energy system dominated by renewables, but is still inconsistent on the potential of offshore storage. The literature on offshore storage spans project-level techno-economic analyses and large-scale energy system models. These strands of literature operate at different spatial scales and lead to heterogeneous conclusions.

1.2.1 Mixed evidence on storage siting across different contexts

Outcomes from single case studies point in multiple directions regarding the siting of storage. Bijl (2019) shows that offshore batteries can reduce curtailment and improve project economics in a Dutch case, while Qazi et al. (2025) find offshore batteries profitable in hybrid North Sea parks due to congestion reduction and stabilising services, but also note capital costs of offshore infrastructure might be underestimated. An analyses of Portuguese offshore parks, including batteries and hydrogen, also find positive results for offshore storage. However, they also note that profitability highly depends on cost assumptions and pipeline expansion (Varotto et al., 2025). Juhlin et al. (2024) directly compare a German offshore wind farm with subsea hydro-pneumatic storage technology to onshore batteries. They find that both are preferable compared to no storage. Moreover, subsea hydro-pneumatic storage is found to economically outperform an onshore battery, but this is due to its longer lifetime and no degradation rather than the siting choice (Juhlin, Eren, et al., 2024).

At the same time, several case-based analyses highlight that offshore flexibility options

are not universally attractive. Van den Haak (2023) finds that when land availability and grid access are unconstrained, onshore storage remains more cost-effective. Similarly, Atherton et al. (2023) examine forty-seven Swedish wind farms, but find no suggestion that shifting storage offshore would lead to added benefit. More broadly, reviews of offshore energy storage concepts highlight that marine foundations, installation logistics, harsh operating conditions, and complex maintenance requirements substantially increase capital and operational expenditures relative to onshore alternatives (Santos et al., 2022; Wang et al., 2019).

These non-uniform outcomes demonstrate that neither onshore nor offshore storage is intrinsically superior. Their relative value depends significantly on assumptions made, local grid condition, and the balance between infrastructure costs and avoided constraints. While this project-level literature is essential for parameterizing models and understanding economic barriers, it does not establish whether offshore storage is cost-effective from a Dutch system perspective. Cost-effectiveness is often assessed in isolation, without endogenous competition against alternative flexibility, onshore storage, or network solutions. As a result, conclusions drawn from net present value (NPV) or cost-benefit analyses cannot directly answer under what system configurations offshore storage would be selected in a cost-minimal energy system, particularly with a high degree of renewable penetration.

1.2.2 System-level and spatially detailed modelling approaches

At the system level, Wiegner (2025) demonstrates the importance of storage to achieve deep decarbonization and a reliable North Sea energy system. Wiegner’s model describes the value of short-, medium-, and long-term storage technologies for balancing renewables with a comprehensive sensitivity analysis of 2,000 energy system configurations. This approach provides interesting conceptual insights about the systemic conditions under which offshore or onshore storage is preferred. However, the model is high-level on a few important aspects. It represents just one offshore and one onshore node without landing point complexity, makes general distance assumptions (50–300km), and does not use the high shares of offshore winds expected in the Netherlands. As a result, outcomes are strongly driven by cost assumptions and backup availability (Wiegner, 2025) and cannot represent the spatial and infrastructural challenges that determine the actual value of different storage locations in the Netherlands.

At the other end of the spectrum, detailed spatial models that account for infrastructure, such as Bos (2025), highlight how storage can relieve congestion when sited at provincial nodes near offshore wind landing zones. These works show that storage siting matters, and that transmission expansion alone cannot eliminate all challenges. However, they remain limited to onshore batteries, and therefore overlook the potential role of offshore storage.

1.2.3 Knowledge gap

The literature thus exhibits a methodological gap. Project-level studies lack system optimisation and substitution effects, while system-level models often lack spatial detail and ignore the complex offshore-onshore interface. Spatially detailed grid models, in turn, rarely include offshore storage as a candidate technology. As a result, offshore storage has not been evaluated within a spatially resolved, cost-optimising national model that explicitly represents constraints at export cables, landing points, and the onshore grid, while allowing endogenous competition between alternative flexibility options.

1.3 Research questions

This thesis addresses this gap by assessing the system-level role of offshore electricity storage in a future Dutch power system with high offshore wind capacity. The analysis focuses on identifying the conditions under which offshore storage contributes to system value, defined primarily as a reduction in total system cost while maintaining reliability constraints. The main research question is formulated as follows:

Main question

Under which system conditions does offshore electricity storage reduce total system costs in the 2040 Dutch power system with increased offshore wind capacity?

To address this question, the research is structured around three sub-questions:

Sub-questions

- **SQ1.** Which conditions determine the relative system-level value of offshore electricity storage compared to onshore alternatives in the Dutch power system?
- **SQ2.** Under which combinations of the identified conditions (SQ1) does the cost-minimizing model endogenously allocate electricity storage offshore, and to what extent?
- **SQ3.** How does offshore storage deployment affect system costs, infrastructure and technology substitution, and operational performance (curtailment, congestion and emissions) across different scenarios?

Sub-question 1 is addressed in Chapter 2 through the identification of the key conditions. In this thesis, these conditions are defined as structural characteristics of the energy system, such as infrastructure constraints, technology costs, and generation levels, that determine the relative system value of offshore storage compared to onshore alternatives.

Sub-questions 2 and 3 are analysed in the Results chapter, using a scenario-based modelling approach to examine the endogenous allocation of storage and its system-level impacts across scenarios.

2 Condition identification

2.1 Positioning

This thesis adopts a system-level perspective on offshore storage, since the impact of storage on system cost, reliability, and infrastructure is dependent on its interaction with the broader electricity system (Guerra et al., 2016; Krishnan et al., 2016; Wiegner, 2025). Therefore, storage is considered within an infrastructure allocation problem in a cost-optimised power system. In power-system planning and design, infrastructure allocation refers to determining the optimal capacity and location of generation, transmission, and flexibility resources to meet demand at least system cost under specific system constraints (de Oliveira et al., 2025; Guerra et al., 2016; Krishnan et al., 2016). In this perspective, storage is not assessed in isolation but as one of several competing options whose capacity and location are determined within a cost-optimising model.

This perspective implies that the relative system-level value of offshore storage versus onshore alternatives is determined by how each option interacts with constraints across the offshore–onshore system. This chapter therefore identifies the key conditions that shape these interactions, structured along the spatial chain from offshore generation to onshore demand.

The literature is reviewed for general system functions of storage (2.2), how the value of storage becomes location-dependent in offshore-centric systems (2.3), and which technological characteristics shape their feasibility (2.4). Chapter 2 ends with a selection of conditions that determine the relative system-level value of offshore electricity storage compared to onshore alternatives, and with that provides an answer to sub-question 1. This provides the basis for the parametric condition experiments conducted in Chapter 4 (Results).

2.2 General system functions of storage in high-renewable systems

The first function through which storage contributes to system value is temporal reallocation of electricity. By shifting energy from periods of surplus to periods of scarcity, storage reduces short-term supply–demand mismatches. It thereby lowers reliance on fuel based dispatchable generation capacity. In planning models, this is reflected in avoided investments in peaking assets and fuel cost, provided reliability constraints are satisfied (Denholm et al., 2010; Zakeri & Syri, 2015).

A related second function is improved renewable generation utilisation. In high-renewable systems, excess production may exceed demand or network transfer capacity, leading to curtailment. Storage can absorb part of this surplus, increasing utilisation of generation assets and altering optimal investment decisions. Importantly, curtailment is not inherently inefficient in cost-optimised systems. Storage is only deployed when its cost is lower than the value of otherwise curtailed energy or alternative flexibility measures (Hirth, 2013; Horsch et al., 2018).

The third function is spatial and arises from interaction with transmission infrastructure. Storage can reduce peak flows, defer or substitute network expansion, and change spatial power flows (Krishnan et al., 2016). Conversely, transmission availability shapes storage value by determining where surplus generation can be transported. The system value of

storage is therefore partly location-dependent, emerging from congestion patterns, network topology, and spatial distribution of supply and demand. This spatial dimension becomes more important in a system with high offshore wind capacity and a geographical mismatch between supply and demand.

In planning models that co-optimize generation and transmission, storage competes directly with network reinforcement, dispatchable generation, and imports. It is deployed only when it offers a lower-cost way of meeting demand and reliability constraints than these alternatives (Guerra et al., 2016). As a result, its system value is inherently context-dependent and shaped by network constraints, alternative flexibility options, and the spatial distribution of supply and demand (Krishnan et al., 2016; Zerrahn et al., 2017). Again, in offshore-centric systems where generation is geographically separated from demand, and constrained by export and landing infrastructure, these interactions become decisive for storage siting.

2.3 Determinants of storage siting in systems with large offshore wind shares

In systems with large shares of offshore wind, generation is spatially separated from demand and much of the transmission network. As a result, the system value of storage becomes location-dependent, as offshore and onshore storage interact differently with export cables, landing infrastructure, inland congestion, and alternative flexibility options (Krishnan et al., 2016; McKenna et al., 2021; Singlitico et al., 2021; Zerrahn et al., 2017). This section therefore identifies the conditions that change the relative system value of offshore storage vs onshore alternatives.

2.3.1 Upstream conditions

Offshore export cable sizing and wind overplanting

Offshore wind farms deliver electricity to shore via offshore substations and export cables. If export cables are built with a capacity equal to the rated capacity of the wind park, they are not fully utilised during a substantial share of the year, because wind output varies over time. Given the relatively high capital cost of offshore transmission infrastructure, such significant underutilisation may not be economically optimal. As a result, some wind farms are 'overplanted', meaning that installed wind capacity exceeds export capacity. Although generation must then be curtailed during periods of maximum output, the average cost per installed megawatt can decrease because export infrastructure is used more intensively. Export cable sizing therefore reflects an economic trade-off between higher transmission investment and increased curtailment of wind energy (Li & DeCarolis, 2015; Rabanal et al., 2024).

A growing body of work analyses how offshore storage can change this trade-off. By absorbing surplus production during high-output periods and discharging when export capacity is available, offshore storage can reduce peak export flows and increase effective utilisation of export infrastructure (Kazemi-Robati et al., 2025; Li & DeCarolis, 2015; Wiegner et al., 2024). This smoothing effect can make overplanting more economically attractive, by limiting curtailment and increasing the amount of exported energy (Atherton et al., 2023; Bijl, 2019). In this case, the location of storage is essential. Congestion at the export cable occurs upstream of the onshore grid, meaning that storage located onshore and connected downstream cannot directly relieve constraints on the export cables themselves. Offshore storage can, because it acts before the constrained export cables.

This condition is not specific to the Dutch power system. In any offshore wind system, export capacity, wind overplanting, curtailment, and storage form an interdependent design problem. Whether offshore deployment is beneficial depends on relative costs and system constraints, but the structural trade-off itself is universal.

Hybrid interconnectors

In addition to domestic connection, offshore wind generation may be exported to neighbouring countries through interconnectors. Recent policy developments favour so-called hybrid interconnectors, in which cross-border transmission cables are integrated with offshore wind infrastructure rather than constructed as point-to-point connections (Ministerie van Klimaat en Groene Groei, 2025a). Such hybrid designs allow export of wind power during high-output periods and import of electricity when offshore generation is low, thereby providing an alternative to curtailment (Rettig et al., 2023; Singlitico et al., 2021).

The availability of alternative export routes moderates the export-cable trade-off described above. If offshore wind can be routed to foreign markets, pressure on domestic landing capacity is reduced, lowering the marginal value of offshore storage. However, projected hybrid interconnection capacity is much lower than the ambitions for offshore wind capacity (Ministerie van Klimaat en Groene Groei, 2025a). The moderating effect will thus not materialize at the majority of wind parks. Another important note is that interconnection shifts surplus energy spatially to other markets, whereas storage shifts it temporally within the domestic system. Consequently, additional interconnection may relieve congestion but does not necessarily increase the availability of renewable electricity for domestic consumption.

Co-location versus stand-alone infrastructure

The relative attractiveness of offshore and onshore storage is further influenced by infrastructure-sharing opportunities. Offshore storage would be co-located with wind farms and connected to existing offshore substations and export cables. In such configurations, storage shares grid connection infrastructure, which reduces the need for additional infrastructure investments (Li & DeCarolis, 2015; Wiegner et al., 2024).

Onshore storage can also be co-located, for example at renewable generation sites or industrial facilities, allowing (partial) use of existing grid connections. However, onshore storage can also be a stand-alone asset, requiring new grid connections and potentially local reinforcement. This decreases the relative system-level value of stand-alone onshore storage projects, but this is no structural distinction for all onshore storage.

2.3.2 Offshore-onshore interface conditions

Landing point capacity and onshore spatial constraints

The rapid expansion of offshore wind towards 30–50 GW by 2040 comes with integration challenges at the coast (Ministerie van Klimaat en Groene Groei, 2025a). As stated in the Wind Energy Infrastructure Plan, *'Until 2040, the rollout of offshore wind energy will mainly take place via electrical connections'* (Ministerie van Klimaat en Groene Groei, 2025a). Policy documents indicate that electrical landing capacity is expected to grow at least to 30 GW in 2040, with a potential extension towards approximately 40 GW, provided that sufficient spatial and technical conditions can be realised (Ministerie van Klimaat en Groene Groei, 2025b). The VAWOZ 2031–2040 programme is explicitly tasked with identifying feasible landing points under increasing spatial pressure (Ministerie van Economische Zaken

en Klimaat, 2022). A key uncertainty could be whether suitable landing points can be developed at the required pace and scale, rather than realising offshore potential.

At sea, export infrastructure is comparatively modular. The standardised 2 GW HVDC concept functions as the basic building block for new connections (Ministerie van Economische Zaken en Klimaat, 2022). Scaling offshore export capacity largely entails adding additional radial systems. The documents do not identify fundamental technological limits to offshore cable capacity itself. Constraints at this level are primarily financial and logistical rather than physical.

The coastal interface introduces more pronounced uncertainties. Both PAWOZ (focused on Eemshaven) and VAWOZ emphasise the sensitivity of the Wadden Sea and other Natura 2000 areas, crossings of primary flood defences, complex dynamics in coastal zones, and competing land uses (Ministerie van Economische Zaken en Klimaat, 2022; Ministerie van Infrastructuur en Waterstaat, 2022; Ministerie van Klimaat en Groene Groei, 2025b). While complicating factors, they are largely route-specific constraints, but do not necessarily impose an absolute upper bound on landing capacity.

The structurally more binding constraint emerges onshore at the level of converter stations and 380 kV substations. Electrical connection requires a converter or transformer station (for a 2 GW HVDC link typically occupying several hectares) and a connection to a 380 kV station with sufficient available capacity (or new station) (Ministerie van Economische Zaken en Klimaat, 2022). Importantly, there are practical limits to the amount of offshore wind that can be injected at a single 380 kV station, related to thermal loading, voltage stability, and system security requirements (indicatively ~ 6 GW) (Ministerie van Economische Zaken en Klimaat, 2022). Many existing 380 kV substations have limited headroom for additional offshore injection, while expanding or constructing new stations requires substantial land take (approximately 15 hectares) and faces spatial constraints in coastal-industrial areas. If sufficient new or expanded landing nodes cannot be realised, the feasibility of reaching approximately 40 GW of electrical landing capacity by 2040 becomes uncertain (Ministerie van Klimaat en Groene Groei, 2025b).

From a system perspective, the dominant landing constraint therefore appears to be the availability, expandability, and system-technical limits of the onshore converter and 380 kV stations. If the binding constraint lies at the level of peak injection into 380 kV nodes, upstream storage may relieve pressure by smoothing instantaneous flows and deferring substation expansion. Upstream storage can both be offshore or onshore at landing points, however, space for onshore storage near landing points is limited. Depending on these spatial constraints, offshore storage can be the only scalable upstream storage option.

2.3.3 Downstream conditions

Transmission constraints

Even if converter stations and 380 kV substations can technically accommodate additional offshore connections, inland transmission corridors connected to these landing nodes may become congested under high wind output. Large peak injections from offshore wind can require reinforcement of existing high-voltage lines or the construction of new transmission assets. Such reinforcements are technically feasible but capital-intensive, time-consuming, and subject to permitting and spatial constraints, as recognised in national infrastructure planning programmes (Ministerie van Economische Zaken en Klimaat, 2022; Ministerie van Klimaat en Groene Groei, 2025b).

Upstream storage can function as an alternative to grid expansion by lowering maximum loading of inland transmission assets. However, as described in the previous section, space

near landing points is limited. Offshore storage can thus become relatively more attractive if onshore storage near landing points is constrained or downstream grid expansion is relatively expensive.

2.3.4 Competing pathways (onshore storage, grid expansion, interconnection, curtailment)

The individual conditions described above do not uniquely favour offshore storage. In a cost-minimising system, congestion at the export, landing, or inland transmission level can be addressed through several competing responses. These include deployment of onshore storage (particularly near landing nodes), reinforcement of transmission infrastructure, expansion of interconnection capacity, or acceptance of higher curtailment.

Offshore storage therefore does not emerge simply because constraints exist, but only when it provides a lower system cost or more spatially feasible response than these alternatives. Its relative system-level value is thus determined by the comparative costs and scalability of these alternatives, in interaction with the upstream, interface, and downstream conditions described above.

2.3.5 System conditions

Offshore CAPEX and OPEX premium

Offshore electricity storage is subject to structurally higher capital and operational expenditures than comparable onshore installations. First, marine foundations, offshore installation logistics, and more complex maintenance increase both upfront investment and lifetime operating costs. Moreover, onshore storage is already a mature technology, with much lower risk and uncertainty than offshore storage. From a system perspective, this implies that offshore storage must generate sufficiently high location-specific benefits to compensate for its cost premium (IRENA, 2025; Sepulveda et al., 2018).

Offshore wind capacity

The relative value of offshore storage is also influenced by the penetration level of offshore wind in the generation mix. Scenario studies referenced in the Wind Energy Infrastructure Plan indicate that required offshore wind production capacity in 2040 ranges between 29 and 51 GW, with most scenarios clustering around approximately 40 GW (Ministerie van Klimaat en Groene Groei, 2025a). As the share of electricity produced offshore increases, a larger fraction of system variability and peak injection originates at sea. This amplifies pressure on export cables, landing nodes, and downstream transmission corridors, increasing the likelihood of congestion and curtailment.

Academic studies consistently show that higher shares of variable renewable energy increase the need for flexibility and raise the marginal value of storage, particularly when transmission expansion is constrained (Denholm et al., 2010; Sepulveda et al., 2018; Zakeri & Syri, 2015). In offshore-centric systems, this effect is spatially concentrated at the sea-land interface.

2.4 Technology dimension

2.4.1 Offshore storage technologies

The offshore storage literature identifies several candidate technologies, which differ fundamentally in discharge duration, maturity, and spatial constraints.

Electrochemical storage (lithium-ion batteries)

Lithium-ion battery energy storage systems (BESS) currently dominate onshore grid-scale deployments. They are modular, fast-responding, and typically designed for multi-hour discharge durations. In the offshore context, they can be installed at turbine level, on AC platforms, connected to HVDC links, or embedded within converter stations. A recent systematic review of storage provision in HVDC-connected offshore wind farms concludes that BESS is the most versatile and promising technology among those assessed (Rabanal et al., 2024). The authors attribute this to three structural properties.

First, BESS offers a balanced power–energy profile: it combines fast response with sufficient energy capacity to deliver peak shaving, curtailment mitigation, DC support, and secondary control, unlike long-duration storage (slow response) or ultra-fast technologies (limited energy). Second, it is locationally flexible, deployable onshore or offshore and across AC and DC infrastructure without geological constraints. Third, lithium-ion BESS is technologically and commercially mature, with well-documented costs, efficiencies ($\sim 85\text{--}90\%$), and lifetimes (~ 15 years), reducing parameter uncertainty relative to alternative offshore storage concepts (Rabanal et al., 2024; Santos et al., 2022).

Subsea hydraulic storage (offshore pumped hydro)

Subsea pumped hydro concepts, such as the StEnSea system developed by Fraunhofer IEE (Fraunhofer Institute for Energy Economics and Energy System Technology (IEE), 2025), exploit hydrostatic pressure at depth to enable energy storage on the seabed. Recent engineering analyses emphasise lessons from offshore oil and gas infrastructure and provide indicative cost frameworks for deepwater subsea storage structures (Juhlin, Slocum, & Assadi, 2024). These systems target larger energy capacities than batteries and are conceptually suited to multi-hour or multi-day shifting.

However, their economics are dominated by subsea civil works, installation logistics, and site-specific constraints. Technology readiness remains below that of lithium-ion BESS, and parametrisation in national-scale capacity-expansion models would require substantial speculative assumptions.

Underwater compressed air energy storage

Underwater CAES concepts use rigid tanks or flexible bags at depth to exploit ambient hydrostatic pressure for near-isobaric storage. Reviews of offshore CAES options demonstrate technical feasibility at prototype scale. Similar to subsea pumped hydro, these systems aim at medium- to long-duration storage. Yet performance, efficiency, and cost remain design-dependent, and offshore deployment is still at an experimental stage (Vella et al., 2017; Wang et al., 2015).

Hydrogen

Offshore hydrogen production is often discussed as an alternative to electrical export. Strictly speaking, hydrogen is not electricity storage but carrier conversion. However, when export constraints are structural rather than peak-driven, shifting energy from electricity to hydrogen could relieve bottlenecks more effectively than short-duration batteries (Travaglini et al., 2025). At the same time, hydrogen deployment is characterised by substantial uncertainty regarding electrolyser costs, infrastructure build-out, demand development, and regulatory frameworks. These uncertainties complicate parametrisation in long-term system models (Ramakrishnan et al., 2024).

2.4.2 Technology scoping

The offshore storage literature does not identify a universally optimal technology. Rather, it shows that storage value depends on service type, discharge duration, and its position relative to export and landing constraints. Still, as near term scalable solution, most studies favor BESS deployment.

This thesis deliberately narrows the scope to offshore lithium-ion battery energy storage. The objective is not to assess all offshore storage technologies, but to isolate a specific system question: under what conditions does (short-duration) offshore electricity storage create structural value in a wind-dominated system with limited export capacity?

Lithium-ion is selected because it occupies a balanced position between power and energy capability, allowing fast response while retaining multi-hour shifting potential. Its technological maturity enables defensible parametrisation of efficiency, cost, and lifetime, and its modularity makes deployment at offshore hubs technically plausible. Importantly, lithium-ion can be deployed both offshore and onshore with broadly comparable technical characteristics. This allows the analysis to focus on locational effects rather than on fundamentally different storage physics. In contrast, technologies such as subsea pumped hydro or underwater CAES would introduce offshore-specific characteristics that are not directly comparable to their onshore counterparts, risking results driven by technology effects rather than siting effects.

This technological restriction implies that the analysis primarily addresses peak-driven congestion and short-term flexibility. Structural energy imbalance, where average offshore generation persistently exceeds export capability, may instead favour carrier-conversion pathways such as hydrogen, but fall outside the storage representation adopted here.

2.5 Synthesis: condition framework and research implications

2.5.1 Spatial structure and condition interaction

The conditions discussed in Section 2.3 reveal that offshore-centric power systems exhibit a layered spatial structure between generation and demand (see Figure 2.1). As offshore wind capacity increases, a larger share of system variability and peak injection originates at sea and must pass sequentially through export cables, landing nodes, and inland transmission corridors before reaching demand. Each of these layers is governed by distinct technical, spatial, and economic constraints that influence how energy flows can be accommodated.

Storage interacts differently with each layer depending on its location. Offshore storage affects flows before they encounter export and landing limits. Storage located at or near landing nodes interacts with converter and substation capacities, but the availability is limited due to spatial constraints. The system-level effectiveness of storage therefore depends on where in the spatial chain constraints become binding and how storage at that location interacts with them.

Storage siting is thus not a binary offshore vs onshore distinction, but a question of how flexibility is positioned within the spatial architecture of the system. The allocation outcome depends on which layers become limiting under increasing offshore wind penetration and on how different response options interact with those limits.

2.5.2 Selection of dominant conditions

The preceding sections identified a set of conditions influencing the relative system-level value of offshore storage versus onshore alternatives. Not all conditions, however, warrant

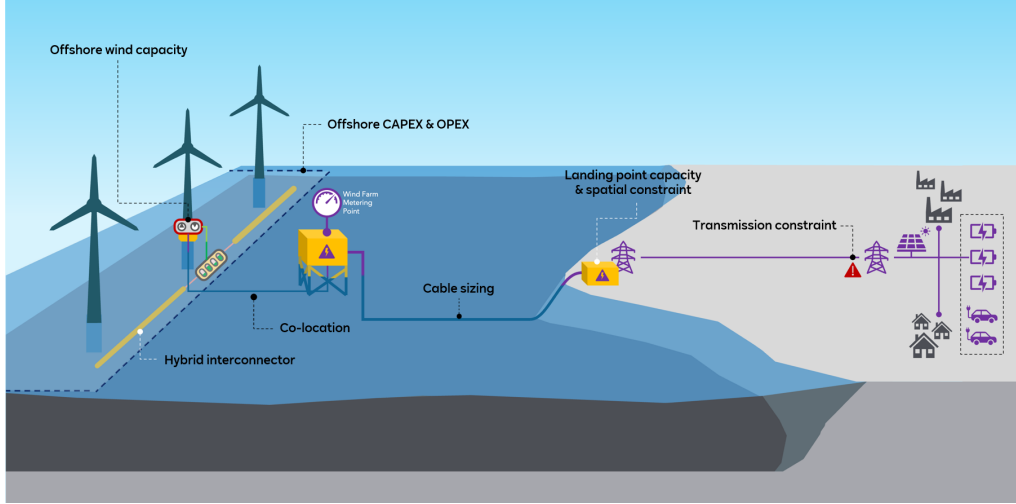


Figure 2.1: Illustration of condition interactions - Adapted from FLASC (2024)

explicit variation in the experimental design. To delimit the scenario space while preserving analytical relevance, conditions are selected for scenario variation when they satisfy three conditions:

1. Locational leverage: the condition can alter the location in the spatial network where constraints become binding.
2. System-wide impact: the condition affects total system cost (and endogenous infrastructure allocation).
3. Deep uncertainty in 2040: the condition is subject to substantial policy, cost, or deployment uncertainty.

The 3 conditions that score highest on these criteria are later included in scenario analysis. The other conditions are either included in the model in another way or abstracted from the model. The conditions are displayed in table 2.1 and discussed below.

Scenario varied conditions

Offshore wind capacity:

Offshore wind capacity directly scales upstream injection and determines the degree of pressure on export cables, landing nodes, and inland transmission corridors. Higher penetration increases the likelihood that upstream or interface constraints become binding and thus materially influences the spatial value of storage. Given the uncertainty in the policy range of approximately 30–50 GW by 2040, offshore wind capacity satisfies all three selection criteria and qualifies for scenario variation.

Onshore near-landing storage availability:

The availability of storage capacity near landing nodes determines whether upstream congestion can be relieved onshore or whether offshore storage becomes the only scalable upstream option. This condition directly affects the spatial substitutability between offshore and onshore storage and therefore has strong locational leverage. Its future availability is uncertain due to unclear landing point locations, spatial constraints and competing land use, justifying explicit variation.

Offshore storage CAPEX and OPEX premium:

Offshore storage is characterised by higher installation and maintenance costs relative to

onshore technologies, combined with substantial uncertainty due to limited deployment experience. The cost differential directly determines whether offshore storage can compete with onshore storage and transmission expansion in a cost-minimising system. Given its high impact and deep uncertainty, it is treated as a scenario condition.

Secondary conditions

Landing point capacity:

Landing capacity is a central interface constraint because it determines how much offshore generation can be injected into the onshore high-voltage grid at a limited set of coastal nodes. For the offshore–onshore storage trade-off, the decisive factor is not landing capacity in isolation, but its tightness relative to offshore wind capacity. A given landing envelope may be non-binding at low offshore wind penetration and strongly binding at high penetration; conversely, higher landing capacity can neutralise otherwise binding offshore build-out.

For this reason, the analysis can treat landing capacity as a secondary condition while varying offshore wind capacity, since the system effect is primarily governed by their ratio and interaction (i.e., whether offshore build-out pushes the interface into a binding regime). In the experimental design, landing point capacity is represented as an existing baseline with limited expansion potential rather than as an independent scenario axis.

Export-cable sizing and overplanting:

Upstream of the landing point, export cable capacity is a distinct constraint. As described in 2.3.2, this export infrastructure is comparatively modular and does not seem to be a structural bottleneck, making it less deeply uncertain than other conditions. Export capacity is therefore represented as a baseline level with expansion possible at a cost, allowing the model to endogenously choose between additional export capacity, curtailment, and offshore storage. It is not varied as an independent scenario axis. (Curtailment is treated as an endogenous operational outcome rather than a scenario driver).

Inland transmission expansion:

Transmission reinforcement is modelled as an endogenous response with associated investment costs. Since it constitutes a competing system response rather than an exogenous constraint, it is not varied as a scenario parameter. Imposing exogenous transmission limits would predetermine substitution outcomes and reduce analytical neutrality.

Hybrid interconnection:

Projected hybrid interconnection capacity remains limited relative to total offshore wind ambitions and does not fundamentally alter the spatial constraints within the domestic system. It is therefore included as expansion possible at a cost, bounded by interconnector projections. This again allows the model to endogenously choose between additional export capacity, curtailment, and offshore storage

Co-location versus stand-alone infrastructure:

Differences between co-located and stand-alone storage primarily affect connection costs at a detailed spatial level. Given the lack of consistent nationwide data and the limited structural impact on where constraints are binding, this condition is not included in the model.

Condition	Locational leverage	Impact	Uncertainty	Treatment
Offshore wind capacity	High	High	High	Varied
Onshore near-landing storage availability	High	High	Medium–High	Varied
Offshore storage CAPEX/OPEX premium	Medium–High	High	High	Varied
Landing point capacity	High	High	High	Bounded*
Inland transmission expansion	High	High	Low (endogenous)	Endogenous
Export-cable sizing / overplanting	Medium–High	High	Medium	Endogenous
Hybrid interconnection	Low–Medium	Medium	Low–Medium	Endogenous
Co-location vs stand-alone connection costs	Low	Low–Medium	High (local)	Abstracted

* Baseline + costed expansion; binding regime determined by interaction with offshore wind capacity.

Table 2.1: Assessment of conditions based on locational leverage, system impact, and uncertainty.

Together, the selected conditions capture the main uncertainties that determine where constraints bind in an offshore-centric system and whether flexibility is placed offshore or onshore. The experimental design therefore varies offshore wind capacity, landing point storage availability, and the offshore cost premium, while allowing export, landing, transmission, and interconnection expansion to respond endogenously within defined planning limits. Chapter 3 translates this framework into a spatially resolved capacity-expansion model.

3 Methodology

3.1 Methodological approach

This chapter describes the modelling approach used to assess the system-level role of offshore electricity storage in the Dutch power system in 2040. The analysis uses a spatially resolved techno-economic optimisation model in Calliope that minimises total system cost under different system conditions. This thesis defines total system cost as the sum of annualised CAPEX, fixed and variable OPEX, fuel costs, CO2 costs, and network expansion costs across the full temporal and spatial scope of the model. Total system cost is used as the operationalisation of system value throughout this analysis. This choice follows directly from the research objective, assessing cost-effectiveness, and from the modelling framework, in which cost is the only metric that consistently integrates investment, dispatch, and infrastructure decisions. In addition, demand must be met at all times, ensuring system reliability through model constraints. Storage, generation, transmission, and selected offshore infrastructure options are endogenously optimised within predefined technical and spatial bounds.

The literature review identified the main conditions expected to shape the relative value of offshore storage compared to onshore alternatives (SQ1), including offshore wind build-out, near-landing onshore storage availability, and the offshore cost premium. These conditions are translated into a structured scenario design, while other system responses, such as transmission expansion, curtailment, and technology substitution, are determined endogenously by the model.

The model is used for two analytical purposes. First, it is used to identify under which conditions the cost-minimising system allocates storage offshore. Second, it is used to evaluate how offshore storage affects total system cost, infrastructure requirements, and operational outcomes relative to alternative system configurations.

3.2 Model description

3.2.1 Framework and system boundary

The analysis is carried out with Calliope, an open-source energy system modelling framework for techno-economic optimisation. Calliope formulates the energy system as a mixed integer linear optimisation problem in which investment and dispatch decisions are jointly optimised subject to technical, spatial, and temporal constraints (Pfenninger & Pickering, 2018). The objective is to minimise total annualised system cost.

In this thesis, the model is limited to the electricity system. It includes electricity generation, storage, transmission, and cross-border interconnection, but excludes detailed modelling of other energy carriers and sectors. The spatial scope is the Dutch electricity system, represented by offshore wind areas, landing points, inland nodes, and international interconnectors. The temporal scope is a 2040 future system, using hourly variation in demand and renewable generation profiles.

Most technology assumptions, demand levels, and availability constraints are based on the NBNL Energy Transition Model *Koersvaste Middenweg* scenario. This scenario is characterized by rapid electrification, based on current trends and policy ambitions (TenneT,

2025a). This scenario is thus used as the general system background. An existing Calliope model using this scenario by Hasselaar (2026) is adapted and extended. This thesis modifies the electricity network and technology representation, and adds offshore nodes, landing points, offshore transmission links, and offshore storage options.

3.2.2 Spatial configuration

The model represents the Dutch electricity system as a network of offshore nodes, landing point nodes, inland nodes, and foreign nodes (only those relevant for interconnection). Inland nodes represent the high-voltage switching and transformer substations present or under construction in 2025. Landing point nodes represent the coastal interface where offshore electricity can come ashore, i.e. DC to AC converter stations. These are based on present, planned (until 2030) and possible (until 2040) landing locations (Ministerie van Economische Zaken en Klimaat, 2022, 2026; Ministerie van Klimaat en Groene Groei, 2025b). Offshore nodes represent the main offshore wind areas considered in Dutch North Sea planning. Again, these are based on present, planned (until 2030) and possible (until 2040) offshore windparks (Ministerie van Economische Zaken en Klimaat, 2024, 2026; Ministerie van Klimaat en Groene Groei, 2025a). Not all wind farms are modelled individually. To keep the model tractable while preserving the main spatial structure of the offshore system, multiple wind farms are clustered using their coordinates and (possible) export cable routes, to be represented by one node (see Appendix). For instance, Nederwiek IA, Nederwiek IB, Nederwiek II and Nederwiek III are clustered to 1 node. In total, this results in 6 offshore wind hubs.

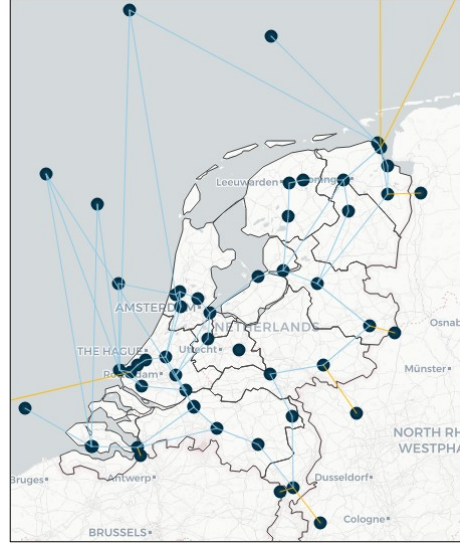


Figure 3.1: The modelled transmission network

The nodes are connected via edges, representing high-voltage lines. The onshore edges represent the 51 bi-directional high-voltage transmission lines connecting onshore substations, consisting of 38 lines operational in The Netherlands in 2025, three lines under construction and operational before 2040, one line in planning, and nine interconnectors connecting to the Dutch grid to neighbouring grids (TenneT, n.d.-a). The offshore edges represent one-directional high-voltage (DC) transmission lines connecting wind hubs with landing points. These are based on 'promising solution directions for electric cable routes in the North Sea' (Ministerie van Economische Zaken en Klimaat, 2026). Finally, edges between landing points and inland nodes represent the connection between converter stations and the onshore 380 kV network. These links are used to model both landing point capacity constraints and the costs of landing point build-out.

3.2.3 Technology and infrastructure representation

Technologies

The model includes the main electricity generation, storage, and flexibility technologies required to analyse the role of offshore electricity storage in the Dutch 2040 power system. Technologies represented at the nodes include variable renewable electricity generation (onshore wind, offshore wind, rooftop PV, utility-scale PV, and run-of-river hydropower), conventional dispatchable generation (nuclear, CCGT, OCGT, biomass, and waste incinera-

tion), and lithium-ion battery storage (onshore and offshore). The model is solved at hourly resolution for the year 2040, using node-specific hourly demand profiles, renewable generation profiles, and interconnector import/export prices derived from the NBNL *Koersvaste Middenweg* scenario dataset.

Most technology assumptions, including technical performance, existing capacities, and availability bounds, are based on the NBNL Energy Transition Model scenario *Koersvaste Middenweg*. In general, lower bounds are based on 2030 projections and upper bounds on 2050 projections, with an additional margin to preserve optimisation freedom. This thesis adjusts some technologies and bounds that are directly relevant to the offshore storage research question or limit optimisation freedom, which are described below.

First, offshore wind is represented explicitly at offshore hub nodes rather than being connected directly to the onshore system. The total offshore wind capacity range is based on Dutch policy ambitions for 2040, with a scenario range of 30–50 GW. Technology and cost assumptions for offshore wind are taken from the OESTER project (TNO, 2025).

Second, offshore lithium-ion battery storage is added as a separate technology option at offshore hub nodes. Its techno-economic representation is based on the same general battery characteristics as onshore lithium-ion storage, but with an additional offshore cost premium. This premium is constructed based on offshore-specific infrastructure cost estimates for battery storage from TNO and North Sea Energy studies, and is varied between 25% and 100% across scenarios (Jepma et al., 2018; Verstraten & van Dooren, 2025).

Third, several onshore assumptions are adjusted relative to the original NBNL dataset. For onshore lithium-ion batteries, lower bounds per node are reduced to include only large realised or under-construction projects, rather than target values (Energy Storage NL, n.d. TenneT, n.d.-b). This avoids imposing battery deployment levels exogenously at nodes and preserves model freedom to endogenously allocate storage. At the same time, the upper bound is increased to include other types of battery storage, such as home batteries and vanadium batteries. In addition, battery cost assumptions are revised using TenneT’s *Monitor Leveringszekerheid*, allowing separate representation of the power and energy cost components (TenneT TSO B.V., 2025). For utility-scale and rooftop PV, the lower bound is adjusted downward relative to the NBNL scenario to better reflect more recent projections by PBL rather than target values (Planbureau voor de Leefomgeving, 2025). Finally, lower bounds on gas-fired power plants are set to zero, allowing the model to endogenously phase out gas capacity if this is cost-optimal. The final parameter values used in the model are reported in table 3.1.

Infrastructure

Transmission infrastructure is represented in two ways. Offshore export links connect offshore hubs to landing points, while inland transmission links connect landing points to the onshore network and the onshore network itself. This separation is necessary because it allows the model to distinguish between constraints offshore, at landing points, and further inland. Landing point build-out is not modelled as a separate asset class. Instead, its capacity limits and associated costs are represented on the inland links between landing points and inland substations.

A distinction is made between HVAC and HVDC transmission. HVAC is used for onshore transmission lines and interconnectors, while HVDC is used for most offshore export cables. Onshore cable routes are based on the current grid and grid planning (Hoogspanningsnet.com, n.d. TenneT, 2024). Offshore cable routes and length are based on potential routes in VAWOZ, combined with coordinates of wind hubs and landing points, and a common cable

routing factor of 1.15 to approximate more realistic length (Hines et al., 2025; Ministerie van Economische Zaken en Klimaat, 2024; Noordzeeloket, n.d.).

Transmission losses are represented using a distance-dependent efficiency parameter, which is different for HVAC and HVDC (U.S. Energy Information Administration, 2018). For HVDC, conversion losses are also included.

Transmission costs are also distinct for onshore HVAC and offshore HVDC cables. They are represented using cost parameters per unit of transmission capacity and distance, based on a transmission infrastructure cost review (L’Abbate, 2022). For HVDC, also offshore platform and converter costs are included in the export cable, based on a review paper on investment model cost parameters for transmission infrastructure (Härtel et al., 2017). Onshore converter costs are included in landing point build-out, represented on the inland link behind a landing point (Härtel et al., 2017). The final infrastructure parameter values are reported in table 3.1.

Technology	Min. (2030) installed Capacity [GW]	Max. (2050) + 25% installed Capacity [GW]	Fixed Cost		Variable Cost		Technical Characteristics		
			CAPEX [Mj/GW]	WACC	OM annual [Mj/GW/yr]	OPEX [t/MWh]	Efficiency (one-way)	Emissions [tCO ₂ /MWh]	Lifetime [years]
BESS onshore	2,9	93,7	300 + 200/GWh	4%	2%	n/a	92,20%	n/a	15
BESS offshore	0	-	450 + 300/GWh	4%	2%	n/a	92,20%	n/a	15
Offshore Wind	30	50	2500	4%	65,6	n/a	n/a	n/a	20
Onshore Wind	7,4	23,5	1392	4%	17,3	n/a	n/a	n/a	20
PV Rooftop	26,1	170,6	961	4%	12,6	n/a	n/a	n/a	25
PV Utility	6,9	49,6	644	4%	11,8	n/a	n/a	n/a	25
Run-of-River Hydropower	0,04	0,04	3600	4%	32,0	n/a	n/a	n/a	45
Biomass Standalone	0,4	0,5	1400	4%	20,0	5,3	n/a	n/a	40
Waste Incinerator	0,4	0,5	4500	4%	0,2	1,2	n/a	0,54	20
CCGT Gas	0*	15,7	750	4%	11,3	2,5	60,0%	0,34	30
OCGT Gas	0*	5,1	749	4%	7,5	3,8	36,0%	0,52	40
Landing points	30	40	81	3%	2%	n/a	n/a	n/a	n/a
Transmission HVDC	30	50	174 + 2.6/km	3%	2%	n/a	99,997%/km - 1,3%*	n/a	30
Transmission HVAC	135	159,3	0.4/km	3%	2%	n/a	99,993%/km	n/a	30
Interconnector	12,6	12,6	0.4/km	3%	2%	n/a	99,993%/km	n/a	30

Demand [TWh]
278,7
ETS price [€/tCO ₂]
177

*Lower bounds for gas plants were set to 0 to allow for complete phase-out.
*A 1.3% loss is applicable to every HVDC cable due to AC - DC conversion

Table 3.1: Parameter values per technology

3.3 Scenario design

The scenario design focuses on the main conditions identified in Chapter 2 as most relevant for the system-level value of offshore electricity storage. Rather than varying all uncertain inputs, the analysis concentrates on a limited number of key dimensions, while other variables are treated as endogenous within predefined bounds (see previous section).

Three scenario dimensions are varied:

- Offshore wind capacity $\in \{30, 37.5, 45\}$ GW
- Near-landing onshore storage capacity (per node) $\in \{0, 560, \text{unconstrained}\}$ MWh
- Offshore storage CAPEX/OPEX premium $\in \{25\%, 50\%, 100\%\}$

First, offshore wind capacity is varied to represent different levels of system scale and pressure on export cables and landing points. The upper bound of offshore wind (45 GW) exceeds the maximum landing point capacity allowed in the model (40 GW), reflecting

landing points as a key bottleneck identified in Chapter 2. Export cable capacity can expand up to 50 GW, while existing landing and export capacity is set at 30 GW. Under these conditions, the model can respond through a combination of overplanting, curtailment, infrastructure expansion, and storage deployment. The spatial distribution and expansion limits of offshore wind, export cables, and landing points are based on recent Dutch policy and planning documents (Ministerie van Economische Zaken en Klimaat, 2024, 2026; Ministerie van Klimaat en Groene Groei, 2025a, 2025b).

Second, the availability of onshore battery storage near landing points is varied. This reflects uncertainty in whether storage can be deployed at locations where it most directly competes with offshore storage. Three levels are considered: no availability, limited availability (560 MWh per node), and unconstrained deployment. The intermediate level represents a stylised spatial constraint in which battery deployment occupies approximately the same area as the converter station itself (around 4 ha) (Ministerie van Economische Zaken en Klimaat, 2022; Repowered & APPM, 2023). For comparison, this corresponds to roughly 40% of the largest battery project currently under construction in the Netherlands (approximately 1400 MWh) (Energy Storage NL, n.d.).

Third, the cost of offshore battery storage is varied through a premium relative to onshore lithium-ion storage. This reflects uncertainty in the additional costs of offshore installation and infrastructure. Estimates based on TNO assumptions suggest a premium in the order of 33–47% (Verstraten & van Dooren, 2025), while estimates derived from North Sea Energy studies suggest a wider range of approximately 45–106% (Jepma et al., 2018). As these estimates remain uncertain and do not yet account for potential cost reductions from alternative integration concepts such as battery integration in monopile structures (TNO, 2025), a range of 25–100% is applied in this study.

For each scenario, two model runs are performed: one with offshore storage enabled and one with offshore storage disabled. This allows isolating the system-level contribution of offshore storage. The full scenario design is shown in table 3.2.

Scenario Design — Conditions Overview		
Condition	Levels (varied across scenarios)	Rationale
C1 — Offshore wind capacity Lower bound $W_{\min}$ [GW]	30 GW	Spans the policy range of 30–50 GW by 2040; 45 GW serves as a stress scenario. Upper bound fixed at 50 GW across all runs.
	37.5 GW	
	45 GW	
C2 — Near-landing onshore storage Capacity upper bound [MWh]	0 MWh	0 MWh reflects severe spatial constraints at landing nodes; 560 MWh reflects partial availability (similar land use as station); spatially unconstrained.
	560 MWh	
	Unconstrained	
C3 — Offshore storage CAPEX/OPEX premium Relative to onshore equivalent	+25%	Derived from offshore infrastructure cost estimates (TNO, North Sea Energy). Range partly depending on energy/capacity ratio: +25%; +50%; +100%.
	+50%	
	+100%	
Offshore storage availability Applied within each scenario	Enabled	The disabled run provides the system (cost) contribution of offshore storage (relevant to SQ3).
	Disabled	

Total: $3 \times 3 \times 3 = 27$ scenarios $\times$ 2 runs (enabled/disabled) = 54 model runs.

Table 3.2: Scenario design

3.4 Experiments and evaluation metrics

Model results are analysed to answer sub-questions SQ2 and SQ3 using two complementary analytical approaches and a structured set of key performance indicators (KPIs). The KPIs

are organised by sub-question and divided into primary indicators that directly answer the sub-question, and secondary indicators which can provide further explanation.

Analytical approach

For SQ2, the endogenous allocation of storage is examined across the full scenario space to identify conditions under which offshore storage is selected by the cost-minimising model. For each of the 27 scenarios, the model determines the cost-optimal distribution of all technologies. The presence, location, and scale of offshore storage deployment are compared across scenarios to identify which combinations of offshore wind capacity, near-landing storage availability, and offshore cost premium lead to offshore storage being part of the optimal configuration.

For SQ3, a different comparison is used. For each scenario that is found to include offshore storage, results from runs with offshore storage enabled are compared to runs in which offshore storage is disabled. This isolates the contribution of offshore storage to system performance and allows identifying which system components it substitutes or complements.

KPIs

The KPIs used to evaluate model results are summarised in Table 3.3 and described below.

SQ2 — Allocation indicators

The primary allocation indicator is total installed offshore storage capacity [GWh and GW]. This directly answers whether offshore storage is selected and at what scale. A secondary indicator is offshore storage utilisation [full-load hours per year], which provides insight into actual usage. Another secondary indicator is the spatial distribution of storage allocation across offshore hubs, which could be compared to storage capacity at landing points and inland nodes.

SQ3 — System impact indicators

The primary system-level indicator is total system cost difference [M€/year], calculated as the difference in annualised total system cost between the offshore-storage-enabled and offshore-storage-disabled runs within each scenario. This is the primary measure of system value as operationalised in this thesis.

Secondary substitution indicators capture how offshore storage affects competing technologies and infrastructure. These include changes in export cable capacity expansion [GW], indicating whether offshore storage substitutes for offshore HVDC investment; landing point capacity expansion [GW], reflecting whether storage defers coastal interface build-out; and inland transmission expansion [Meur], showing whether downstream grid reinforcement is alleviated. In addition, changes in onshore storage deployment [GW/GWh], disaggregated by node type (landing point versus inland), indicate whether offshore and onshore storage act as substitutes. Finally, changes in other flexible generation capacity, mainly gas, and import/export [GW and GWh] capture the effect on dispatchable backup requirements.

Secondary operational indicators capture how offshore storage changes system behaviour. Offshore wind curtailment [GWh/year and % of potential generation], provides the most direct measure of whether upstream export constraints are relieved. Landing point utilisation [% of hours above 95% rated capacity] serves as a proxy for whether the coastal interface remains binding with and without offshore storage. Similarly, inland congestion [% of hours above 95% rated capacity] provides insights for the onshore grid. Lastly, system CO₂ emissions [MtCO₂/year] are included as a secondary indicator.

KPI	Unit	Sub-question	Type
Installed offshore storage capacity	GW / GWh	SQ2	Primary
Offshore storage utilisation	FLH/yr	SQ2	Secondary
Spatial distribution of storage	Node-level GWh	SQ2	Secondary
Total system cost difference	M€/yr	SQ3	Primary
Export cable capacity change	GW	SQ3	Secondary
Landing point capacity change	GW	SQ3	Secondary
Inland transmission change	GW·km	SQ3	Secondary
Onshore storage deployment change	GWh (by node type)	SQ3	Secondary
Flexible generation change	GW / GWh	SQ3	Secondary
Import/export change	GWh	SQ3	Secondary
Offshore wind curtailment	GWh / %	SQ3	Secondary
Landing point utilisation	% hrs > 95% capacity	SQ3	Secondary
Onshore congestion	% hrs > 95% capacity	SQ3	Secondary
System CO ₂ emissions	MtCO ₂ /yr	SQ3	Secondary

Table 3.3: Key performance indicators used in the analysis

Optional expansion — Near-optimal exploration

Within each scenario, near-optimal system configurations are explored.

- For each scenario, near-optimal alternatives are generated using MGA/SPORES with an optimality slack of $\varepsilon = 2.5\%$, producing N alternative system designs.
- These alternatives are used to evaluate the flexibility of system design within the same external conditions.
- Node-level recurrence of offshore storage deployment is computed to identify locations providing structural system value. Locations are classified based on recurrence frequency:
 - **No-regret:** $> 90\%$ occurrence,
 - **Robust:** 50–90% occurrence,
 - **Considerable:** $< 50\%$ occurrence.

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Appendices

Appendix A